

(see chapter 2). Let's normalize W_t to be one so we don't have to worry about it. The portfolio returns, $r_{p,t+1}$, in equations (4.6) and (4.7) depend on the portfolio weights chosen at the beginning of the period, x_t , as equation (4.1) emphasizes. Thus, we can write equation (4.7) as

$$\begin{aligned} E[U(W_{t+5})] &= U(W_t)E[U(1 + r_{p,t+1}(x_t))U(1 + r_{p,t+2}(x_{t+1})) \dots U(1 + r_{p,t+5}(x_{t+4}))] \\ &= E[U(1 + r_{p,t+1}(x_t))U(1 + r_{p,t+2}(x_{t+1})) \dots U(1 + r_{p,t+5}(x_{t+4}))]. \end{aligned} \quad (4.8)$$

Figure 4.2 sketches an outline of the dynamic programming solution. Let's start at the end, at $t+4$ to $t+5$, where the investor chooses portfolio weights to maximize expected utility at the terminal horizon $T = t + 5$. This is Panel A of Figure 4.2. This is a static one-period problem, and, for CRRA utility without

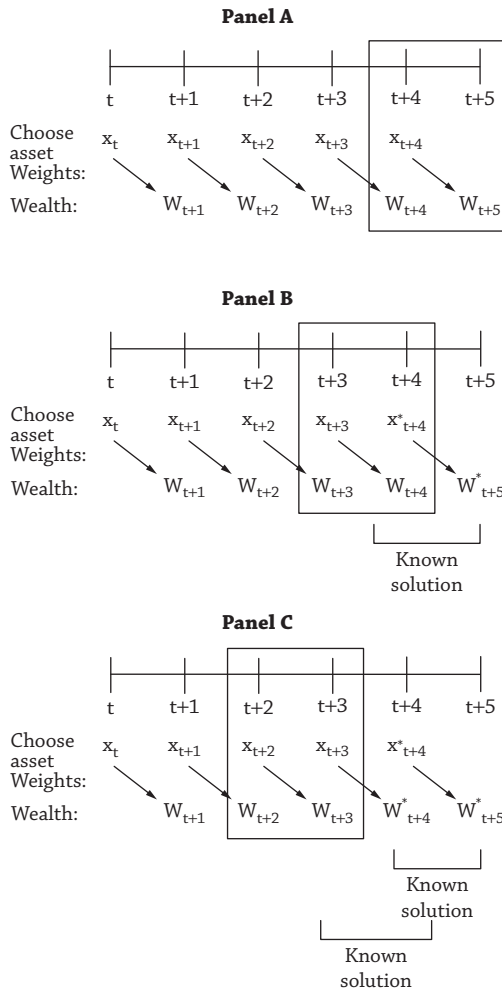


Figure 4.2